Using deep learning to generate synthetic tax records

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The Budget Lab at Yale

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What do we want?

Augmented tax unit microdata
for every year, including future years

What do we have?

- **One year of tax data, filers only**
  - About ten years out of date
  - Some values are blurred and some records are distorted
  - No non-filers, and none of the other variables a tax calculation needs: consumption, wealth, unrealized gains, estates
- **Surveys** that hold some of this information, but noisily and not at administrative quality
- **Aggregate targets**
  - Past years: IRS totals by AGI group, filing status and age range
  - Future years: CBO-style projections of 1040 items, or of total wages, business profits and similar totals

The task

Turn these scattered sources into

synthetic, realistic, augmented tax unit microdata

for every year

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains
Filers (PUF)	Married	812	$142k	$310	$5.2k
	Single	1,404	$38k	$0	$0
	Married	655	$0	$4.9k	$61k
	Head of hh.	1,120	$51k	$12	$0
	Single	3	$2.3m	$88k	$3.9m

1. **Start with the IRS PUF**, whatever year is available.

Made-up records.

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains
Billionaires	Married	1	$1.2m	$14m	$310m
Filers (PUF)	Married	812	$142k	$310	$5.2k
	Single	1,404	$38k	$0	$0
	Married	655	$0	$4.9k	$61k
	Head of hh.	1,120	$51k	$12	$0
	Single	3	$2.3m	$88k	$3.9m
Non-filers	Single	2,310	$0	$0	$0
	Single	1,870	$6.5k	$0	$0

2. **Add records**: non-filers, and the very rich the PUF leaves out.

Made-up records.

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains	Net worth	Consump-tion	Tips
Billionaires	Married	1	$1.2m	$14m	$310m	$4.1b	$6m	$0
Filers (PUF)	Married	812	$142k	$310	$5.2k	$610k	$96k	$0
	Single	1,404	$38k	$0	$0	$9k	$34k	$6.1k
	Married	655	$0	$4.9k	$61k	$2.4m	$88k	$0
	Head of hh.	1,120	$51k	$12	$0	$31k	$47k	$0
	Single	3	$2.3m	$88k	$3.9m	$38m	$410k	$0
Non-filers	Single	2,310	$0	$0	$0	$2k	$19k	$0
	Single	1,870	$6.5k	$0	$0	$1k	$15k	$1.2k

3. Add columns: estimate Y given X in a survey, where X is also on the PUF; draw Y for each record.

Made-up records.

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains	Net worth	Consump-tion	Tips
Billionaires	Married	1	$1.2m	$14m	$310m	$4.1b	$6m	$0
Filers (PUF)	Married	860 ↑	$142k	$310	$5.2k	$610k	$96k	$0
	Single	1,290 ↓	$38k	$0	$0	$9k	$34k	$6.1k
	Married	702 ↑	$0	$4.9k	$61k	$2.4m	$88k	$0
	Head of hh.	1,180 ↑	$51k	$12	$0	$31k	$47k	$0
	Single	2 ↓	$2.3m	$88k	$3.9m	$38m	$410k	$0
Non-filers	Single	2,450 ↑	$0	$0	$0	$2k	$19k	$0
	Single	1,760 ↓	$6.5k	$0	$0	$1k	$15k	$1.2k

4. **Reweight to the target year**: grow weights for population and dollars for prices, then move each weight up or down until the year's totals are hit.

Made-up records.

What were the problems with this approach?

- **It did not work well for past years.** The affordability project made this clear.
- **A separate model for each variable**, each estimating its conditional relationships in its own way.
- **Reweighting worked for only a few past years.** Tax law changed what the IRS targets measure, mainly TCJA cutting back itemized deductions.
- **Stretched further every year.** Plausible records for future years, but worse with each year away from 2015.

The ideal approach

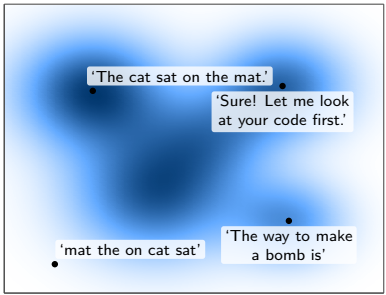
1. Take in survey data covering the joint relationships among every variable we might want to model.
2. Learn all of those relationships in **One Huge Joint Probability Distribution**.
3. For any target year, supply the targets that describe that year, and pull out of it the records that match them.

The result: a realistic synthetic microdata set for that year.

From a conversation with John and Harris.

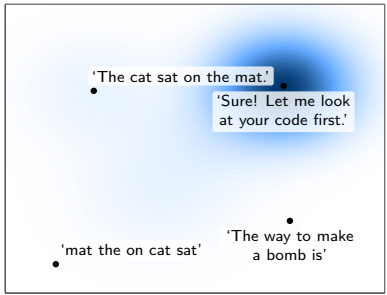
Language models: pretraining, then post-training

Pretraining



Read the whole internet and learn which sentences are likely: a prior over language

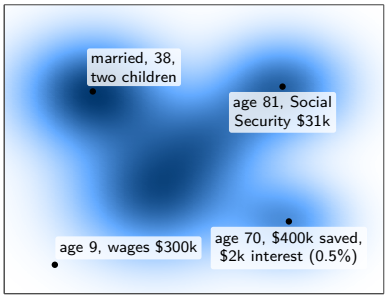
Post-training



Squeeze the mass toward a helpful, harmless assistant, and away from nonsense and harm

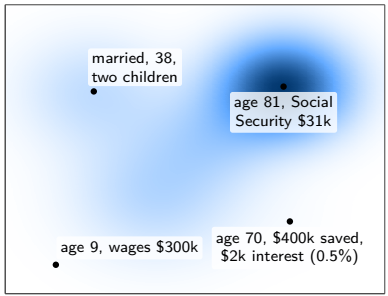
Our model: pretraining, then post-training

Pretraining



Learn which records are likely, from every source: a prior over records

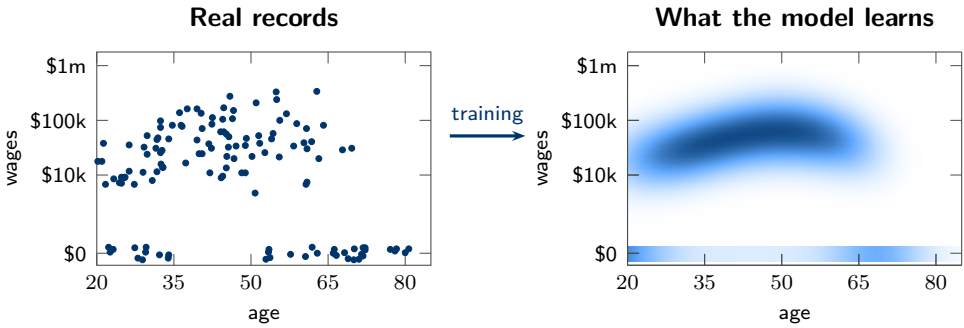
Post-training for 2040



Squeeze the mass toward records that hit 2040's totals, each still coherent

Post-training finds the distribution closest to the prior that hits the year's totals.

Training: learn the shape of the data

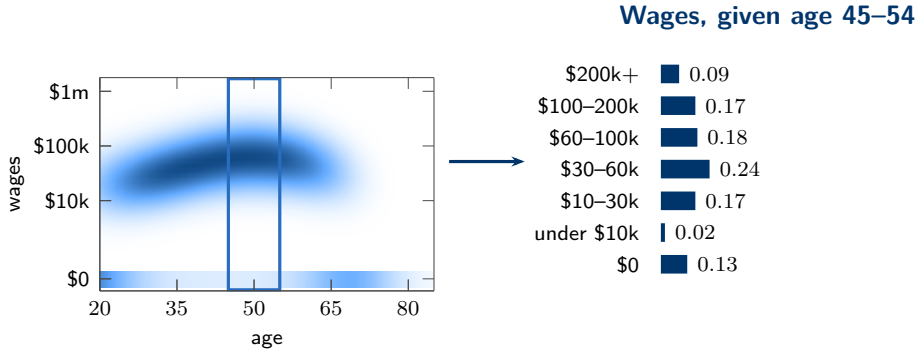


Hide some of a real record's values; the model guesses them from the rest and is nudged toward the truth. Repeat many times over every record.

It learns how likely each combination of values is: two variables here, all at once in practice.

Illustration.

Asking the model a question: take a slice

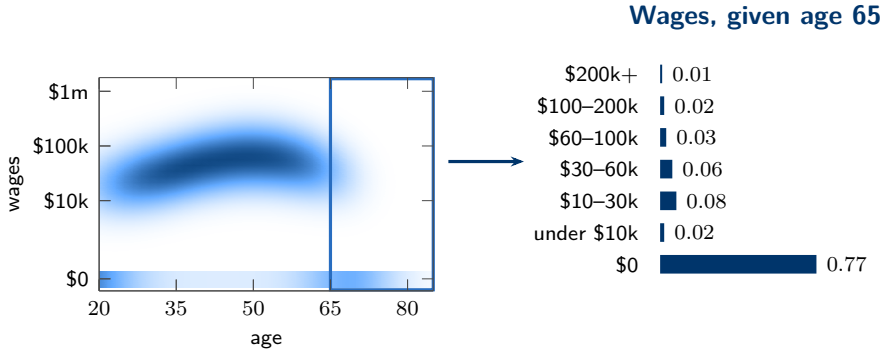


Any question of the form "this variable, given those": wages given age, age given wages, net worth given age and wages.

The answer is a probability for each range. Every dollar amount has 84 ranges: exactly zero, 10 for losses, 73 for positive amounts (finer at the top).

Illustration. Seven ranges shown here.

Asking the model a question: take a slice

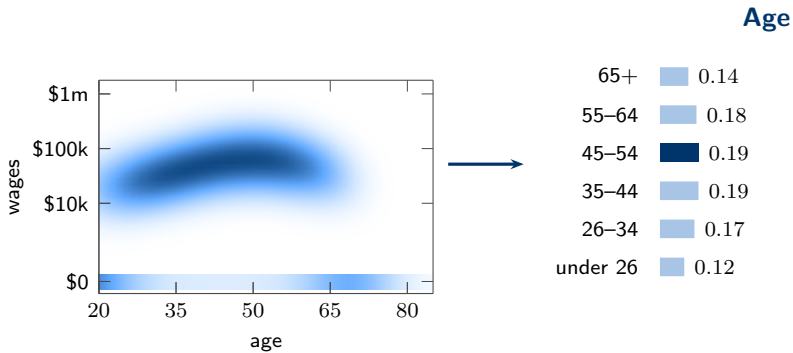


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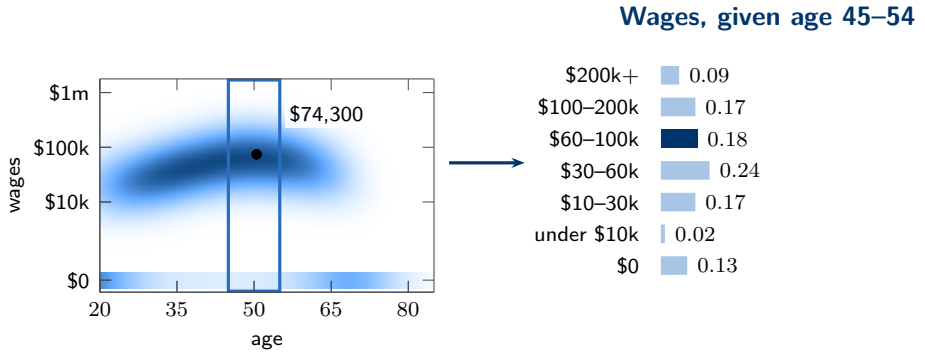
Writing a record: draw, take the slice, draw again



Draw the first variable from its probabilities: age 45–54.

Illustration.

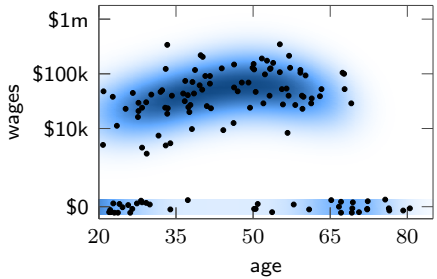
Writing a record: draw, take the slice, draw again



Draw wages from the slice at that age: \$60–100k. A real amount from that range becomes the record's dollars: \$74,300.

Illustration.

Writing a record: draw, take the slice, draw again



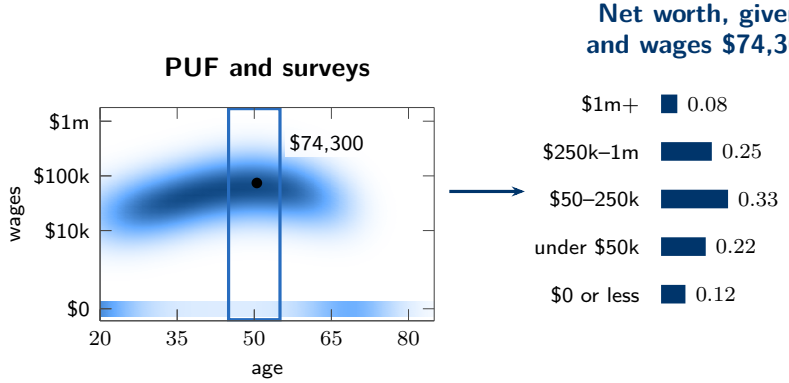
130 records drawn this way.

They land where the data are dense, zeros included.

Each draw uses everything drawn so far. The tax variables come in a random order; in the real run, 400,000 filer records, each started from its AGI group.

Illustration.

A source without a variable still helps

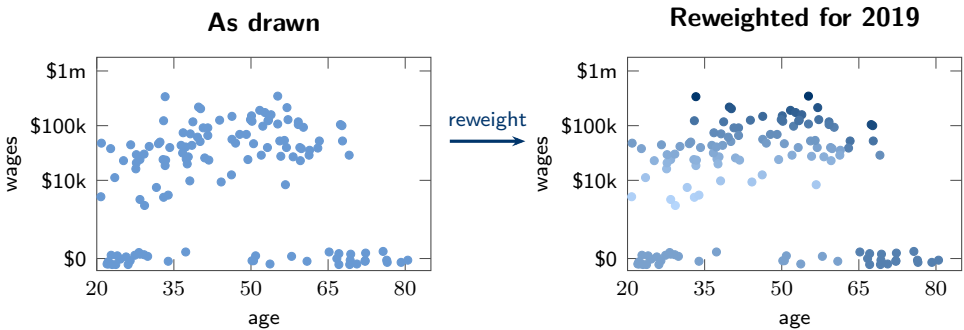


The PUF has no net worth and is never asked for it, but it shapes the age and wage picture.

In a drawn record, survey-only variables come after the tax ones, answered as the SCF would see the household.

Illustration.

Post-training: reweight the drawn records

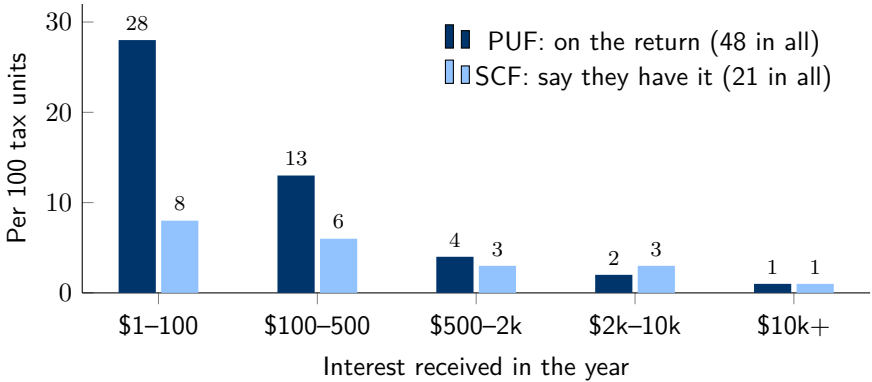


Each record keeps every value; only its weight changes (darker = more weight).

The smallest change in weights that brings each of 2019's 753 published totals within its standard error (half of one for the SCF's wealth totals).

Illustration.

Challenge: surveys and returns measure differently



The gap is all in small amounts: they go unreported in the survey. Above $2,000 the two sources agree.

Married, wages $75–200k. 2015 PUF and 2016 SCF (tax year 2015), weighted.

The fix: two slots and a fill-in

SCF household	interest, as said	interest, as on a return
household 1	$250	$250
household 2	$0	?
household 3	$0	?
household 4	$1,800	$1,800

The answer given is kept. A second slot holds what the return would show, the only slot the PUF has.

Per 100 couples	PUF	SCF	Fill in
$1–100	28	8	20
$100–500	13	6	7
$500–2k	4	3	2
$2k and over	3	3	0
Any	48	21	28

The PUF sets how many are filled in, and at what sizes. The rest of the SCF record sets which households.

Married, wages $75–200k. 2015 PUF and 2016 SCF (tax year 2015), weighted. Rows may not add because of rounding.

Other challenges

- **Building tax units from surveys.** Surveys count households. Each one is split into tax units by the same rules for every survey, and then we decide who files.
- **The return itself changes.** TCJA cut back itemizing; new items, like the tips and overtime deductions, are on no past return.
- **A thin top.** Few very rich records in any source.
- **Future years.** No microdata, only a handful of projected totals.
- **Checking the result.** No source has every variable, so some relationships have nothing to check against.

Where we are

- A 2019 file: 400,000 filer records, plus non-filers
- Five sources: the 2015 PUF, the SCF, the CEX, the CPS and the SIPP, every survey split into tax units by the same rules
- Every PUF item, plus a balance sheet, consumption, tips and auto loan interest
- About 750 targets, all met within their standard errors: IRS tables by AGI, filing status and age, the top 1%, 0.1% and 0.01%, W-2 tips, SCF wealth

What is left

- Rules applied after drawing: credit eligibility, death probabilities, unrealized gains, the basis of capital gains
- The file Tax-Simulator reads, and a first run of Tax-Simulator on it
- A tax calculator inside post-training, for targets on computed tax
- Other years: 2015 to 2023, then the projection years
- The very top: the PUF's aggregate records and the Forbes 400

Questions